Noncooperative Game Theory: from Reasoning Patterns to Solution Concepts

Zhenran Guo*

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Preface

This is my note on noncooperative game theory. After reading the core of classical game theory textbooks, e.g., Gibbons (1992a) and Osborne and Rubinstein (1994), I find that organizing the theory merely according to solution concepts—like various refinements of Nash Equilibrium—feels LESS natural, as these texts place limited emphasis on explicitly modeling the players' reasoning patterns. I treat game theory as a general model for understanding how players form reasonable beliefs in different settings of interaction and act rationally in the sense that they choose strategies consistent with their beliefs. Suppose the reasoning pattern is clear, i.e., how players form and update their belief is modeled; then the corresponding solution concept can be defined naturally. Indeed, for almost all important solution concepts, one can find a supporting reasoning pattern, and I believe that understanding this underlying reasoning helps clarify each concept's assumptions and limitations.

Initially, I have a vague idea of rearranging the materials in classical game theory textbooks by defining the best-response on beliefs when I first thought about a unified framework of game theory. I prefer to understanding conventional game theroy with a EGT perspective. My understanding of game theory resembles that described in the introductory part of Brandenburger (2014) and Perea (2012). Besides, I also refer to (copy) classical game theory textbooks (Fudenberg and Tirole (1991), Osborne and Rubinstein (1994), Myerson (1997), Gibbons (1992b) and Gibbons (1992a)) on definitions, results and proofs.

*See my homepage for updates. I deliberately use "his/her" repeatedly when referring to pronouns, even though this may distract the reader (and myself), for fear of the choice of language raising issues of gender inequality or discrimination. In addition, I use "we" throughout the text to show (and as reminder for me) that that these notes are a shared intellectual journey between the author and the reader.

Abstract

This notes on game theory take an epistemic approach, by which we mean that players are modeled to i) form a reasonable belief and ii) maximize his/her payoff according to the belief. There are multiple reasoning patterns and we follow these patterns to define solution concepts. Instead of treating the epistemic foundations formally, we use them as intuition and motivation to unfold the conventional game theory. This notes have 5 main parts concerning different game theoretic settings, in which we first discuss the setting and then move on to different reasoning patterns and the corresponding solution concepts as well as their refinements. To be specific, each part concerns:

(1) Introduction Part: We first define the formal notion for games and then make clear that playing in a game is making decisions under uncertainty. Then we discuss several terminologies commonly used (but less likely to be formally defined), including state, information and knowledge.
(2) Strategic Game: We first discuss the complete information case and then move on to Bayesian games. In each type of games, reasoning patterns are introduced first and then we develop the solution concepts following the reasoning pattern.
(3) Extensive Game with Perfect Information
(4) Extensive Game with Incomplete Information
(5) Topics: This part includes some applications of game theory. We start from multiple benchmark models of sender-receiver game, including cheap talk, signaling game, verifiable model, Bayesian persuasion and model persuasion. Then we move on to the theory of mechanism design, where we first introduce the classical auction theory and take auctions as motivating examples to understand mechanisms.

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Part I

Introduction

In decision theory, we're modeled as acting as if we were maximizing our utility given some exogenous parameters. An example is that producers in competitive markets are price-takers who choose the output that maximizes their profit after receiving the market price. In this setting, no individual producer's behavior affects market structure, and expectations about other producers do not influence decision-making.

However, such perspectives of analysis is overly restriction and the real world is not always the case. For example, collusion requires participants to act based on the expectation of other firms' behavior. This is what game theory studies. Game theory is a subject that deals with interaction among decision-makers, focusing on the existence and properties of equilibrium. Equilibirium is a kind of stability, where no participants have incentive to deviate. We'll formalize this concept later on.

1 Game

1.1 Basic Composition of a Game

Now we start to consider the formal definition of games. We try to formalize these concepts by analyzing the game of rock-scissors-pape.

Example 1.1 (Rock-scissors-pape)**.** In this game, we have 2 players and the consequences can be represented in **table 1**, where 1 stands for win, 0 stands for draw and -1 stands for lose. This is also called the payoff matrix, which is widely used in game theory. The embedding assumption of this game is that players prefer winning to losing, inducing a preference relation.

		Player 2		
		R	S	P
Player 1	R	0, 0	1, -1	-1, 1
	S	-1, 1	0, 0	1, -1
	P	-1, 1	1, -1	0, 0

Table 1: Outcome of the Rock-scissors-pape Game

A game consists of players (along with their preferences) and rules (that maps actions to consequences). Each player i, denoted by A_i, has a set of available actions, called action set $\mathcal{A}_i$. We use a_i to denote a certain action, i.e., $a_i \in \mathcal{A}_i$. As for "rule", it maps the action profile $a = (a_i)_{i\in N}$ to a consequence c, i.e., a mapping g from de Cartesian product of action sets $\mathcal{A} = \prod_{i\in N} \mathcal{A}_i$ to the consequence set C. If we assume the participants are rational, they can surely compare among 2 consequences, inducing a preference $\preceq_i^*$ of player i over C.

Definition 1.1 (Game)**.** A game is a 4-tuple $\langle N, \mathcal{A}, g, \{\preceq_i^*\}_{i\in N}\rangle$, where N is the set of player index. If $\preceq_i^*$ admit a utility representation, we may also write $\langle N, \mathcal{A}, g, \{u_i\}_{i\in N}\rangle$ instead.

Notice that the consequence and the action is associated with a mapping, we can equivalently define the preference relation over $\mathcal{A}$. Let a_{-i} denote the action of other players, i.e., $a_{-i} = (a_j)_{j\in N\setminus\{i\}}$, then the preference of player i can be defined as

$$(a_i^{(1)}, a_{-i}^{(1)}) \preceq_i (a_i^{(2)}, a_{-i}^{(2)}) : \Longleftrightarrow\ g(a_i^{(1)}, a_{-i}^{(1)}) \preceq_i^* g(a_i^{(2)}, a_{-i}^{(2)}).$$

Then a game can be represented by $\langle N, \mathcal{A}, \{\preceq_i\}_{i\in N}\rangle$.

Such a game is of the simplist form since it's almost one-shot and static. We'll introduce history when it comes to the extensive form.

1.2 Actions and Strategies

The above notions are defined on the set of actions, which implicitly assumes that players will only choose one action as their strategies. However, you would never stick to rock when playing the RPS game, in which game you would prefer a randomized play so that the opponents may not guess your next move.

Such a strategy is called a mixed strategy. For the actions you may play, you impose a randomization device on them, i.e., a random variable that tells you which action to choose in this round.

Definition 1.2 (Mixed strategies)**.** A mixed strategy σ_i of player i is a distribution over his/her actiion space, i.e., $\sigma_i \in \Delta(\mathcal{A}_i)$. A pure strategy is a degenerate mixed strategy that assigns probability 1 to a single action.

Then our previous notion fails to work anymore unless we extend the definition of payoff to the set of lotteries over the action space.

1.3 Common Example of Games

We first present some strategic games. The first stream includes some social dillema games:

Battle of the sexes(BoS) or according to Osborne and Rubinstein (1994), Bach or Stravinsky.

The Prisoners Dillema(PD) No equilibrium as a strategic game, but allows coorperation and defection. It describes a scenario where two suspects are put into two cell and if they both confect..

Hawk-Dove, or Chicken Game Similar to PD, but only one coorperation option.

Table 2: Social Dillemas

(a) BoS

	B	S
B	2, 1	0, 0
S	0, 0	1, 2

(b) PD

	D	C
D	3, 3	0, 4
C	0, 4	1, 1

(c) Chicken

	D	H
D	3, 3	1, 4
H	1, 4	0, 0

2 Belief, Information and Knowledge

Playing in a game is making decisions under uncertainty. No matter it's a game with complete information or with incomplete one, players are uncertain about his/her opponents' attitudes towards strategies. In another word, players in a game should predict their opponents' strategies to the best of their knowledge, i.e., a probability distribution. Further, in games with incomplete/imperfect information, players are more or less uncertain about the payoff-relevant state.

To characterize players' uncertainty on the world, we shall introduce some formal concepts. Here, we use the concept of state and type to represent the possible parallel worlds and players are trying to figure out which is the true world.

2.1 State, Signal and Belief Update

We define state as a payoff-relevant element by rewriting the definition of payoff function as $u_i : \mathcal{A} \times \Omega \to \mathbb{R}$, where Ω is the set of all possible state. We understand state as something one can never observe directly, but one can observe some signals (partially) revealing the true state. For example, we can never observe the true value of an artwork, but we can oberve the winning auction price which partially reveals its true value. For a partition P_i of Ω, define a signal function $\tau_i : \Omega \to P_i$ such that $\forall \omega \in \Omega : \omega \in \tau_i(\omega)$. The condition says that the signal is credible, meaning that the signal at least contains the true state.

To characterize the idea that players have some idea about how the true world may be, we define a probability distribution over the state space, i.e., $p_i \in \Delta(\Omega)$. This is called the prior of

player i. When they observe a certain signal, say t_i, they'll update their belief by Bayesian rule. As the signal is credible, we should require the probability of its occurance to be positive, i.e., $\sum_{\omega \in t_i} p_i(\omega) > 0$. If the credibility condition holds, they update their belief as the posterior $\pi_i : P_i \to \Delta(\Omega)$ where

$$\pi_i(\omega|t_i) = \begin{cases} \frac{p_i(\omega)}{\sum_{\omega' \in t_i} p_i(\omega')} & , \omega \in t_i; \\ 0 & ,\text{otherwise.} \end{cases}$$

2.2 From State to Information

2.3 Knowledge and Common Knowledge

2.4 Perfect and Incomplete Information

3 Decision Under Uncertainty

3.1 Lottery and the Induced Preference

3.2 von Neumann-Morgenstern Theorem

3.3 Risk

Part II

Strategic Game

A strategic game is a scenario where players make decisions independently and simultaneously.

To understand the meaning of "independently and simultaneously", let's consider an experimental setting where every subject, anonymously sitting behind a computer, is shown a payoff matrix (he/she may only be able to see his/her own payoff) and asked to choose their strategy. They won't be able to see the outcome until everyone makes their choice. Two important things to understand:

- They don't necessarily to make decisions at exactly the same time.
- It's not necessarily a one-shot game. It can be repeated, but across rounds, no strategic links are allowed.

To make a correct decision, they should care not only about their own choice, but also make decisions under other subjects' possible action, i.e., form a belief and maximize his/her payoff according to this belief. To be more specific, we need to consider two things: how should one interact given opponents' action? And, what are opponents' possible actions? We will formalize this idea soon after we give the definition of a strategic game. When it's the incomplete imformation case, we also have to form a belief on the state and on others' belief on the state.

4 With Perfect Information

In this section, we consider the strategic game with complete information. As discussed before, the game $\langle N, \mathcal{A}, \{u_i\}_{i\in N}\rangle$ is common knowledge to all player.

Before getting into the solution concepts, we make clear what our goal is in strategic games as a player. As we assume players are rational, i.e., payoff seekers, we are solving a maximization problem. Formally, we consider that a player i with the belief $(\mathcal{A}_{-i}, \pi_i)$, where π_i is a probability measure on $\mathcal{A}_{-i}$, is trying to maximize his/her payoff by choosing a strategy $\sigma_i \in \Delta(\mathcal{A}_i)$, i.e.,

$$\max_{\sigma_i\in\Delta(\mathcal{A}_i)} \sum_{(a_i)\in\mathcal{A}} \sigma_i(a_i)\pi_{-i}(a_{-i})u_i(a_i, a_{-i}), \tag{1}$$

and denote its solution the best response correspondence

$$BR_i(\pi_i) := \arg \max_{\sigma_i \in \Delta(\mathcal{A}_i)} \sum_{(a_i) \in \mathcal{A}} \sigma_i(a_i)\pi_{-i}(a_{-i})u_i(a_i, a_{-i}). \tag{2}$$

As long as the belief is reasonable, a game is trivial as player can do their best by playing a strategy in the best-response correspondence. Thus, we'll first formalize the reasoning process.

4.1 Rationality

4.1.1 Individual Rationality and Strictly Dominated Actions

We now begin to understand the very first reasoning pattern. As a rational player, i.e., a payoff seeker, we should only play strategies in best response because it helps us obtain the highest payoff given a specific belief. We say such strategies are rational.

Definition 4.1 (Rationality)**.** We say a strategy $\sigma_i \in \Delta(\mathcal{A}_i)$ is rational if $\exists \pi_i \in \Delta(\mathcal{A}_{-i}) : \sigma_i \in BR_i(\pi_i)$. We say a player is rational if he/she plays a rational strategy.

Well, will our opponents choose their strategies in the same way? It's reasonable to assume that they are rational, which means that they'll NOT choose an irrational strategy. Thus we assign positive probability only to strategies that can be supported by a certain belief. For actions can not be supported by any belief, we call them strictly dominated actions and we'll show that they'll not appear in the support of any mixed strategies.

Definition 4.2 (Strictly dominated action*)**.** We say $a_i \in \mathcal{A}_i$ is strictly dominated if $\forall \pi_i \in \Delta(\mathcal{A}_{-i}) : a_i \notin BR_i(\pi_i)$.

We'll see in the next theorem that a strictly dominated action will not be chosen as a pure strategy since there always exists a mixed strategy that yields a higher expected payoff..

Theorem 4.1 (Pearce)**.** *a_i is strictly dominated $\iff$ there exists a mixed strategy $\sigma_i \in \Delta(\mathcal{A}_i)$ such that*

$$\forall a_{-i} \in \mathcal{A}_{-i} : U_i(\sigma_i, a_{-i}) > u_i(a_i, a_{-i}).$$

Then we say σ_i strictly dominates a_i.

Let NB_i be the set of all strictly dominated actions for player i. Then we can reduce the original game as $\mathcal{G}^1 := \langle N, \mathcal{A}^1, \{\preceq_i^1\}_{i \in N} \rangle$, where $\mathcal{A}^1 = \prod_{i \in N}(\mathcal{A}_i \setminus NB_i)$ and $\preceq_i^1$ is a restrcition of $\preceq_i$ on $\mathcal{A}_i \setminus NB_i$.

* In many textbooks, a strictly dominated action is defined to be strictly dominated by a mixed strategy, which is the condition in **theorem 4.1**. We choose to define in this way to better fit our motivations.

4.1.2 Hierarchies of Beliefs

The above discusses what strategies we believe our oponents will choose and we believe they will choose rational strategies. Will they reason similarly? That is, how should we believe that the oponents believe that we believe they are rational? This kind of belief, referred to as belief of belief, and the extended belief of belief are key to rationalizability.

We introduce the so-called hierachy of belief for player i to capture the above idea:

- first-order belief μ_i^1: belief on other opponents' strategies, i.e., $\mu_i^1 \in \Delta(\mathcal{A}_{-i})$;
- second-order belief μ_i^2: belief on how other opponents believe their opponents will react, i.e., $\mu_i^2 \in \Delta(\mathcal{A}_{-i} \times \prod_{j \neq i} \Delta(\mathcal{A}_{-j}))$;
- third/fourth...infinity-order belief.

We provide a formal definitin.

Definition 4.3 (Hierarchy of Beliefs)**.** Let $X_i^{(1)}$ be $\Delta(\mathcal{A}_{-i})$ and define deductively that $X_i^{(k+1)} := \Delta(\mathcal{A}_{-i} \times \prod_{j \neq i} X_j^{(k)})$. The hierachy of beliefs for player i is a tuple $(\mu_i^1, \mu_i^2, \cdots) \in \prod_{k=1}^{\infty} X_i^{(k)}$. Let the set of hierachies of beliefs for player i be T_i. We say $t_i \in T_i$ is the type of player i.

We say that σ is optimal for a type t_i if it's optimal under the first-order belief of t_i.

Definition 4.4 (Optimal strategies for a type)**.** We say that σ is optimal for a type t_i if $\sigma \in BR_i(\mu_i^1)$.

We can think of a general model of how players choose their strategies and form their belief. Such model is called an epistemic model which is a subjective belief of how the opponents, for every player, it specifies a belief of which strategies his/her opponents play and what their hierachies of belief are.

Definition 4.5 (Epistemic model)**.** Let S_i be the set of player i's strategy. An epistemic model specifies every player's type and a belief of his/her opponents' strategy, i.e., $(t_i)_{i \in N}$ and $(\beta_i)_{i \in N}$, where

$$t_i \in T_i \quad \text{and} \quad \beta_i(t_i) \in \Delta(\prod_{j \neq i} S_i \times T_i).$$

4.1.3 Common Belief in Rationality

The above simply provides a notion to express belief and belief of belief. In this subsubsection, using the epistemic model, we model the common belief in rationality.

First, players believe their opponents act rationally, aka the 1-fold belief in rationality, that is

$$supp(\beta_i(t_i)) \subset \left\{((\sigma_j, t_j))_{j\neq i} \in \prod_{j\neq i} S_i \times T_i : \sigma_j \text{ is optimal for } t_j, \forall j \neq i\right\}. \tag{3}$$

Denote types that satisfy (3) as $\mathcal{R}_i^1$.

Further, for the second-order belief in rationality, we want to express that palyers believe their opponents are 1-fold rational. Thus, we consider types that are consistent with 1-fold belief in rationality. Thus, we require

The idea of rationalizability is based on the reasoning that rationality is a common belief.

Definition 4.6 (k-fold belief in rationality)**.** For an epistemic model, we say it has 1-fold belief in rationality if (3) holds. We say it has k-fold belief in rationality if

We take an arbitary player's view:

(1) first-order rationality: I am rational, so my action must be a best response to some belief. I also believe my opponents are rational, so my first-order belief μ_i^1 assigns probability zero to their strictly dominated actions. This reduces the game from $\mathcal{G}$ to $\mathcal{G}^1$.
(2) second-order rationality: I believe my opponents believe their opponents are rational i.e., they assign probability zero to their opponents' beliefs that assign positive probability to strictly dominated actions.

4.2 Rationalizability

4.2.1 Iterated Elimination of Strictly Dominated Actions

The section above shows how a single player eliminates strictly dominated actions, now we extend it to the game level and assume that players have unlimited reasoning.

By the above motivation, we first eliminate the never-best response by reducing the game into $\mathcal{G}^1 = \langle N, \mathcal{S}^1, \{\preceq_i^1\}_{i\in N}\rangle$, where

$$\mathcal{S}^1 = \prod_{i\in N}\{a_i \in \mathcal{A}_i : \exists \pi_i \in \Delta(\mathcal{A}_{-i}) \text{ s.t. } a_i \in BR_i(\pi_i)\}, \tag{4}$$

and $\preceq_i^1$ is obtained by restricting $\preceq_i$ in $\mathcal{S}^1$. (4) says that players won't take actions in never-best response into account, even as a composition of a mixed strategy, thus player should form a reasonable belief that assigns probability zero to other players' never-best response. Denote the set of reasonable belief for player i as $\mathcal{B}_i^1 := \{\pi \in \Delta(\mathcal{A}_{-i}) : supp(\pi) \subset \mathcal{S}_{-i}^1\}$.

Now we define iteratively, suppose we have defined $\mathcal{G}^1, \cdots, \mathcal{G}^k$ and the set of reasonable

belife is denoted by $\mathcal{B}_i^k$. For each $k \geq 1$, let $\mathcal{S}_i^k$ be the projection of $\mathcal{S}^k$ onto player i's action set, and define the restricted best-response correspondence as

$$BR_i^k(\pi_i) := \arg \max_{\sigma_i \in \Delta(\mathcal{S}_i^k)} \sum_{a_i \in \mathcal{A}_i} \sum_{a_{-i} \in \mathcal{A}_{-i}} \sigma_i(a_i)\pi_i(a_{-i})u_i(a_i, a_{-i}).$$

Thus, define

$$\mathcal{S}^{k+1} := \prod_{i \in N} \{a_i \in \mathcal{A}_i : \exists \pi_i \in \mathcal{B}_i^k \text{ s.t. } a_i \in BR_i^k(\pi_i)\}.$$

The player i will update his/her belief as $\mathcal{B}_i^{k+1} = \{\pi \in \Delta(\mathcal{A}_{-i}) : supp(\pi) \subset \mathcal{S}_{-i}^{k+1}\}$. Finally, by intersecting all $\mathcal{S}^k$, we obtain a set that survives the iterated elimination of strictly dominated actions with belief updates, denoted $\mathcal{S} := \bigcap_{k=1}^{\infty} \mathcal{S}^k$.

4.2.2 Rationalizability

Before giving the formal definition of rationalizability, we provide a result concerning properties of $\mathcal{S}^k$ and $\mathcal{S}$.

Proposition 4.2. *Suppose $\mathcal{S}^k$ is generated as above, then*

(1) $\{\mathcal{S}^k\}$ decreases monotonically;
(2) $\exists K \in \mathbb{N} : \mathcal{S}^K = \mathcal{S}^{K+1} = \cdots = \mathcal{S}$;
(3) $\mathcal{S} \neq \emptyset$;
(4) $\mathcal{S}$ has the best-response property, i.e,

$$\mathcal{S} = \prod_{i \in N} \{a_i \in \mathcal{A}_i : \exists \pi_i \in \Delta(\mathcal{S}_{-i}) \text{ s.t. } a_i \in BR_i(\pi_i)\}. \tag{5}$$

We follow (5) to formalize the definition of rationalizability.

Definition 4.7 (Rationalizability)**.** For a strategic game $\mathcal{G}$, we say $a_i \in \mathcal{A}_i$ is rationalizable if there exists $\{S_i\}_{i \in N}$ such that $a_i \in S_i$ and for any $j \in N$,

$$\forall a_j \in S_j : \exists \pi_j \in \Delta(S_{-j}) \text{ s.t. } a_j \in BR_j(\pi_j).$$

4.3 Equilibrium

In this section, we make clear, given a reasonable belief, how you should behave.

4.3.1 Pure Strategy Nash Equilibrium

4.3.1.1 Definition There're games that people may stick to one single strategies, see the following payoff matrix. No matter it's player 1 or 2, he/she should only choose action A as

		Player 2	
		A	B
Player 1	A	2, 2	0, 0
	B	0, 0	0, 0

Table 3: A Coordination Game

that's the only strategy that gives them a positive payoff. We refer to this simplification of strategy a pure strategy.

Definition 4.8 (Pure Strategy)**.** A pure strategy is a strategy that assigns probability one to a single action.

We assume that every player is palying a pure strategy, then we can simplify (1) and (2) as

$$\max_{a_i \in \mathcal{A}_i} u_i(a_i, a_{-i})$$
$$BR_i(a_{-i}) = \arg\max_{a_i \in \mathcal{A}_i} u_i(a_i, a_{-i}).$$

We refer to the steady state where every subject is playing a best-response equilibrium.

Definition 4.9 (Pure Strategy Nash Equilibrium(PSNE))**.** Denote $BR(a) := \prod_{i \in I} BR_i(a_{-i})$. We say a^* is a PSNE if $a^* \in BR(a^*)$. Or equivalently, a^* is a PSNE if

$$\forall i \in N : (a_i, a^*_{-i}) \preceq (a^*_i, a^*_{-i}), \forall a_i \in \mathcal{A}_i.$$

This leads to an important way of finding the equilibrium:

- STEP I: find the best-response correspondence of each player.
- STEP II: find the action profile satisfying $a^*_i \in BR_i(a^*_{-i})$.

4.3.1.2 Examples BoS; Coordination Game; PD; H-D; Pennis(strictly competitive game)

4.3.1.3 Existence As we've already seen, not every strategic game has a PSNE. In this subsubsection, we provide some conditions that guarentee the existence. By the definition of PSNE, it's natural to use the correspondence version of fixed point theorem. Here we use Kakutani's fixed point theorem, which says that as long as the correspondence is nonempty, closed and convex, there exists a fixed point.

To suffice this condition, we assume the followings:

- Action sets are nonempty compact convex subset of $\mathbb{R}^n$;

- The preference relation is continuous and quasi-concave on $\mathcal{A}_i$.

Then by Kakutani's fixed point theorem, there exists PSNE.

Theorem 4.3. *For a strategic game $\langle N, \mathcal{A}, \{\preceq_i\}_{i\in N}\rangle$, suppose for all $i \in N$, $\mathcal{A}_i$ is nonempty, compact and convex. Further, we suppose $\preceq_i$ is continuous and quasi-concave on $\mathcal{A}_i$, then there exists PSNE.*

4.3.2 Mixed Strategy Equilibrium

For many games, sticking with one strategy is somehow stupid. It's like keep playing scissor in the rock-scissor-pape game, which is clearly unwise. So, players may deliberatly randomize the strategy played.

Other motivating example concerning the introduction of mixed strategy includes the matching pennis, which is similar to the rock-scissor-pape game, and the relationship between auditors and taxpayers, see Osborne and Rubinstein (1994, p. 37).

The idea above is to give the strategies a probability measure, which introduce the randomness of the strategies being played.

Definition 4.10 (Mixed Strategy)**.** Let $\Delta\mathcal{A}_i$ denote the set of probability distribution on $\mathcal{A}_i$. Then $\alpha_i \in \Delta\mathcal{A}_i$ is a mixed strategy.

4.3.3 Correlated Equilibrium

5 With Incomplete Information

Part III

Extensive Game with Perfect Information

6 Definition and Solution Concepts

6.1 Definition

The main difference between strategic game and extensive game is that strategic games ignore the sequence of play. We'll use hitory and player function to depict the sequential play and the corresponding strategies. In this part, we focus on extensive game with perfect information. It's perfect information in the sense that the history realized is common knowledge.

6.1.1 History and Sequence of Play

History records, in the game, the action chosen by players sequentially.

Definition 6.1 (History)**.** Let H be a set of sequences whose component is an action, we say H is a set of history if

(1) $\emptyset \in H$, i.e., initial history is recorded;
(2) $(a_1, \cdots, a_K) \in H \implies \forall k \leq K : (a_1, \cdots, a_k) \in H$;
(3) $\{a_k\}_{k=1}^{\infty} \in H \implies \forall i \in \mathbb{N} : (a_1, \cdots, a_i) \in H$.

We define the following:

Length $l(h)$ is the length of history h, i.e., the length of sequence h.

Terminal Suppose $h_0 = (a_1, \cdots, a_K) \in H$, if there doesn't exist $h = (a_1, \cdots, a_{K+1})$ such that $h \in H$, then we say h_0 is a terminal. Let H^T denote the set of terminal history.

Player Function This helps us to identify who is going to play. Suppose we have history $h_1 = (a_1, \cdots, a_K)$ and $h_2 = (a_1, \cdots, a_{K+1})$, then $P(h_1)$ is the player who takes action after h_1, i.e., $a_{K+1} \in \mathcal{A}_{P(h_1)}$.

Outcome Outcome can be represented by a function $O : H^T \to \mathbb{R}^N$, which is the payoff every player can get.

Definition 6.2 (Extensive Game with Perfect Information)**.** $\mathcal{G} = \langle N, H, P, \{\preceq_i\}_{i\in N}\rangle$ is an extensive game with perfect information where

- N is the set of players;
- H is the set of history;
- P is the player function defined on $H^N := H \setminus H^T$;
- $\preceq_i$ is defiend on the set of outcomes $O := O(H^T)$.

Every realized history is common knowledge to players. If $\max_{h\in H} l(h) < \infty$, then we say the game has finite horizon.

6.1.2 Strategies

Definition 6.3 (Strategy)**.** Define $A(h) := \{a \in \mathcal{A}_{P(h)} : (h, a) \in H\}$, where $h \in H^N$. Player i's stratgy in an extensive game with perfect information is a function $\sigma_i : H^N \cap \{h \in H : P(h) = i\} \to \mathcal{A}_{P(h)}, h \mapsto a \in A(h)$.

Example 6.1 (Strategies in **figure 1a**)**.** By definition:

Player set $N = \{Parent, Child\}$;

Terminal History $H^T = \{\{G\}, \{B, F\}, \{B, P, G\}, \{B, P, B\}\}$

Outcome

Define the function $\Sigma : H^N \to H$ such that

$$\Sigma = \begin{cases} \sigma_{Child}, \text{ if } P(h) = Child \\ \sigma_{Parent}, \text{ if } P(h) = Parent \end{cases},$$

where $\sigma_{Child}(h) = h \times \{Good, Bad\}$ and $\sigma_{Parent}(h) = h \times \{Forgive, Punish\}$.

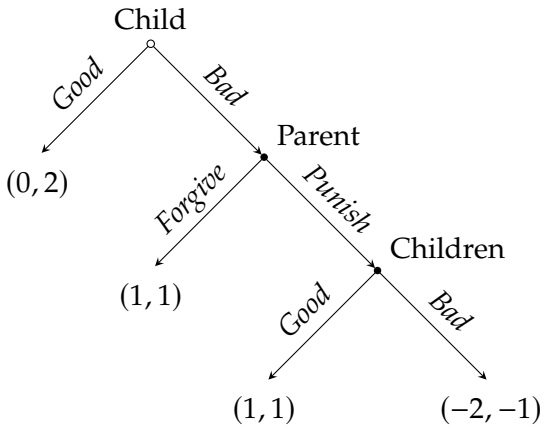
6.1.3 Strategic Form and Reduced Form of an Extensive Game

6.2 Solution Concepts

6.2.1 Nash Equilibrium

This simply treats an extensive game as a strategic game in which the strategies are intepreted as the choice made in a one-shot sense.

Definition 6.4 (Nash Equilibrium(NE))**.**



(a) Extensive Form

	P	F
BB	-2, -1	1, 1
BG	1, 1	1, 1
GG	0, 2	0, 2
GB	0, 2	0, 2

(b) Strategic Form

	P	F
BB	-2, -1	1, 1
BG	1, 1	1, 1
G	0, 2	0, 2

(c) Reduced Form

Note: B = Bad, G = Good, P = Punish, F = Forgive.

Figure 1: Example for extensive form, strategic form and reduced form.

Part IV

Extensive Game with Incomplete Information

Part V

Topics

7 Auction and Introduction to Mechanism Design

This section is largely based on Krishna (2002) and Borgers, Krahmer, and Strausz (2015).

7.1 Single Object Sealed Auction with i.i.d. Private Value

7.1.1 Settings

Suppose there are multiple buyers who bid price for an object. There are several ways to determine who should win the object, among which the most straightforward one is to give the object to the one who would like to pay the highest price. Such an auction is called standard. But, which price should the seller charge? We introduce 2 ways in this subsection.

Before answering the question, we first formalize the setting. Suppose $N = \{1, 2 \cdots, n\}$ is the set of bidders. Player i's monetary valuation on the object is denoted by X_i. Since it's the private value case, we assume X_i's are i.i.d. on an interval $[0, w]$, with a PDF f which has full support. The realization x_i of X_i is private knowledge to player i while the distribution f is common knowledge. Thus we assume common prior here. Finally, we assume bidders has unconstrained budget so that the bidder can pay at most his/her private value to the seller for the object.

Notice that, as long as we specify the rule of object allocation, the payoff is specified as $\mathbf{1}_{\{\text{obtained}\}}(x_i - p)$, where p is the price charged. This is actaully a Bayesian game among bidders. Let $\beta_i : [0, w] \to \mathbb{R}_+ \cup \{0\}$ be the strategy of palyer i and denote $\beta = (\beta_i)_{i \in N}$.

Finally, we assume that the seller/auctioneer has unlimited ability of commitment, i.e., in the ex ante stage, he/she commits to a rule of how the object will be allocated by commiting to a game tree between auctioneer and bidders where the terminal history is assigned a probability distribution over $N \times \mathbb{R}^N$. The distribution describes the probability of a bidder getting the object and the transfer he/she has to pay for the object.

7.1.2 Second-Price Auction

In a second-price auction, the seller gives the object to the one who bids the highest price at the second-highest price. Then the payoff of player i can be formalized as

$$\Pi_i = \mathbf{1}_{\{b_i > \max_{j \neq i} b_j\}}(x_i - p) = \begin{cases} x_i - \max_{j \neq i} b_j & , \text{ if } b_i > \max_{j \neq i} b_j; \\ 0 & , \text{ o.w.} \end{cases}$$

If there are many bidders who bids the same lowest price, the tie is broken by giving them the object at the same probability.

Theorem 7.1 (Truth-telling is weakly-dominant)**.** *$\beta(x) = x$ is a weakly-dominant strategy.*

Proof. W.L.O.G., we consider bidder 1. Let $p := \max_{j\neq 1} b_j$. We prove the case $p > x_1$: for an arbitrary bid $z_i \neq x_1$,

> **Case 1** Suppose $z_1 < x_1 < p$, bidder 1 still loses the object, thus no incentive to deviate.
>
> **Case 2** Suppose $x_1 < z_1 < p$, bidder 1 still loses the object, thus no incentive to deviate.
>
> **Case 3** Suppose $x_1 < z_1 < p$, bidder 1 wins the object with a negative profit $x_1 - p$, thus no incentive to deviate.

Similar argument can be applied to the case of $p \leq x_1$, thus truth telling is a weakly dominant strategy. □

7.1.3 First-Price Auction

In first-price auction, the auctioneer charges the winner the highest bid, thus

$$\Pi_i = \mathbf{1}_{\{b_i > \max_{j\neq i} b_j\}}(x_i - p) = \begin{cases} x_i - b_i & \text{, if } b_i > \max_{j\neq i} b_j; \\ 0 & \text{, o.w.} \end{cases}$$

We formalize how they maximize their profit. Let $Y_i^{(n-1)}$ denote the highest value of i.i.d. r.v. except the ith r.v., i.e., $Y_i^{(n-1)} = \max_{j\neq i} X_j$. Then it's straightforward that it follows the CDF F^{n-1}. Let G denote the CDF. With this notation, we may calculate the (interim) probability that bidder i wins the object, i.e.,

$$\mathbb{P}[\max_{j\neq i} X_j < x_i | X_i = x_i] = \mathbb{P}[Y_i^{(n-1)} < x_i | X_i = x_i] = \mathbb{P}[Y_i^{(n-1)} < x_i] = G(x_i),$$

where the second equation is by the independence. Since we assume bidder i is risk-neutral, we have

$$\beta_i(x_i) \in \arg\max_{b_i \in [0,w]} G(x_i)(x_i - b_i).$$

The solution can be derived by F.O.C.:

$$\frac{d\beta_i^{-1}(b_i)}{db_i} G'\left(\beta_i^{-1}(b_i)\right)(x_i - b_i) - G\left(\beta_i^{-1}(b_i)\right) = 0$$
$$\frac{g\left(\beta_i^{-1}(b_i)\right)(x_i - b_i)}{\beta_i'\left(\beta_i^{-1}(b_i)\right)} - G\left(\beta_i^{-1}(b_i)\right) = 0.$$

Then notice that $x_i = \beta_i^{-1}(b_i)$ and $b_i = \beta_i(x_i)$, we have

$$\begin{aligned} & g(x_i)\left(x_i - \beta_i(x_i)\right) - G(x_i)\beta_i'(x_i) = 0 \\ \iff \quad & g(x_i)\beta_i(x_i) + G(x_i)\beta_i'(x_i) = g(x_i)x_i \\ \iff \quad & \frac{dG(x_i)\beta_i(x_i)}{dx_i} = g(x_i)x_i. \end{aligned}$$

Since $\beta_i(0) = 0$,

$$\begin{aligned} \beta_i(x_i) &= \frac{1}{G(x_i)} \int_0^{x_i} g(y) y dy \\ &= \mathbb{E}\left[Y_i^{(n-1)} \mid Y_i^{(n-1)} < x_i \right]. \end{aligned} \tag{6}$$

This shows that every bidder should bid the above amount. However, notice that it's only sufficient condition for a BNE. Thus, we are left to show that this symmetric strategy is indeed an equilibrium.

Theorem 7.2. *Symmetric BNE of a first-price auction is given in* (6).

Proof. Suppose $b \le \beta_i(w)$ is the strategy bidder i plays in an equilibrium where other bidders bid according to (6). Let $z = \beta_i^{-1}(b)$. The interim expected profit of bidder i bidding b is

$$\begin{aligned} \mathbb{E}\Pi_i(b, x_i) &= G(z)(x_i - b) \\ &= G(z)x_i - \int_0^z g(y) y dy \\ &= G(z)(x_i - z) + \int_0^z G(y) dy. \end{aligned}$$

Meanwhile,

$$\mathbb{E}\Pi_i(\beta_i(x_i), x_i) = G(x_i)(x_i - \beta_i(x_i)) = \int_0^{x_i} G(y) dy. \tag{7}$$

Thus

$$\begin{aligned} \mathbb{E}\Pi_i(\beta_i(x_i), x_i) - \mathbb{E}\Pi_i(b, x_i) &= G(z)(z - x_i) + \int_z^{x_i} G(y) dy \\ &= [G(z) - G(\xi)](z - x_i), \end{aligned}$$

where ξ is between x_i and z. Then $\mathbb{E}\Pi_i(\beta_i(x_i), x_i) \geq \mathbb{E}\Pi_i(b, x_i)$ since G is strictly increasing, which implies $x_i = \beta_i(x_i)$. □

7.1.4 Revenue Equivalence

We are to answer the question which selling mechanism the seller and bidders prefer. We shall compare the ex ante expected payment and revenue, i.e., the expectation of interim expected payment IEP^A and the expected revenue $R^A = n\text{IEP}^A$, where A stands for the form of auction, say $A = I$ for FPA and $A = II$ for SPA. We neglect bidders' strategies here and assume that they choose the equilibrium strategy. Notice that

First-price auction By definition,

$$\text{IEP}^I(x_i) = \mathbb{P}[\text{Win}]\mathbb{E}[\beta_i(x_i) \mid X_i = x_i] = G(x_i)\mathbb{E}\left[Y_i^{(n-1)} \mid Y_i^{(n-1)} < x_i\right].$$

Second-price auction Since bidders reveal the true type in an equilibrium, we have

$$\begin{aligned}\text{IEP}^{II}(x_i) &= \mathbb{P}[\text{Win}]\mathbb{E}[\text{2nd-highest bid} \mid b_i = x_i] \\ &= \mathbb{P}[\text{Win}]\mathbb{E}[\text{2nd-highest value} \mid X_i = x_i] = G(x_i)\mathbb{E}[Y_i^{(n-1)} \mid Y_i^{(n-1)} < x_i].\end{aligned}$$

This ($\text{IEP}^I \equiv \text{IEP}^{II}$) immediately implies that the seller and the bidder should be indifferent between SPA and FPA. Further, we can show that the ex ante expected revenue is the expectation of second-highest price:

$$\begin{aligned}\mathbb{E}\left(R^A\right) &= n\int_0^\omega \int_0^{x_i} y g(y) f(x_i)\, dy dx_i \\ &= n\int_0^\omega \int_y^\omega y g(y) f(x_i)\, dx_i dy \\ &= n\int_0^\omega y g(y)(1 - F(y)) dy.\end{aligned}$$

Also, let $X^{(i)}$ be the ith-highest one of $X_1, \cdots, X_n$, then

$$\begin{aligned}\mathbb{P}\left(X^{(2)} \leqslant y\right) &= \mathbb{P}\left(\left\{i \in N : X_i > y\right\} \leqslant 1\right) \\ &= \mathbb{P}\left(X^{(1)} > y\right) + \mathbb{P}\left(\bigwedge_{i \in N} X_i \leqslant y\right) \\ &= n(1 - F(y))F^{n-1}(y) + F^n(y),\end{aligned}$$

Thus

$$\begin{aligned} f_{X^{(2)}}(y) &= n(n-1)f(y)F^{n-2}(y)(1-F(y)) - nf(y)F^{n-1}(y) + nf(y)F^{n-1}(y) \\ &= n(n-1)f(y)(1-F(y))F^{n-2}(y) \\ &= n(1-F(y))\left(F^{n-1}(y)\right)' \\ &= n(1-F(y))g(y). \end{aligned}$$

This implies

$$\mathbb{E}\left(R^A\right) = \int_0^w y f_{X^{(2)}}(y)dy = \mathbb{E}\left(X^{(2)}\right).$$

One may ask whether this result holds for a wider range of forms of auctions. The answer is yes for standard auctions if we put some restrictions on equilibrium.

Theorem 7.3 (Revenue equivalence principle)**.** *Under the assumption that bidders are risk-neutral and their values are i.i.d., all standard auctions such that 1) the equilibrium is increasing and symmetric and 2) bidders with zero value expect to pay 0 yield the same expected revenue for the auctioneer.*

7.1.5 Reserve Price and Seller's Reasoning

The above implicitly assumes that the seller is risk-neutral so that he/she should be indifferent between SPA and FPA. However, for an auctioneer to decide which auction form to choose, there are two issues to concern:

(1) If he/she is risk-averse/risk-seeking, is there any difference between SPA and FPA?
(2) Even though

The above discusses how the auctioneer would allocate the object with an implicit assumption that his/her monetary valuation of the object is weakly below the ex ante expected profit. However, it's possible that the auctioneer values the object so much that he/she is reluctant to sell it at the expected revenue. We call the lowest price that the auctioneer would rather keep the object the reserve price. There are several ways to make sure that the auctioneer makes profits given some auction form and we formalize some easiest way in this subsubsection.

7.2 Bayesian Mechanism Design: the Private Value Case

In this subsection, we generalize SPA and FPA to a selling mechanism. We still require participants (bidder in auctions) to send some messages and the mechanism designer (auctioneer in auctions) would commit to an allocation rule that assigns messages to an outcome.

Definition 7.1 (Selling mechanism)**.** Let $N = \{1, \cdots, n\}$ be the set of participants. A selling mechanism is a tuple $\langle \mathcal{M}, \pi, \mu \rangle$. It consists of the space of messages $\mathcal{M} = \prod_{i \in N} \mathcal{M}_i$ and an outcome (π, μ) where $\pi : \mathcal{M} \to \Delta$ is the probability of assigning the object to a certain

participant and $\mu : \mathcal{M} \to \mathbb{R}^N$ is the expected transfer participants pay to the mechanism designer for the object.

7.2.1 Auction as a Mechanism and the General Settings

Auction is a mechanism. In a standard auction with i.i.d. private value, the message space is $[0, w]^N$. The auctioneer commit to giving the object to highest bidder(s), i.e.,

$$\pi(b_1, \cdots, b_n) = \left(\frac{\mathbf{1}_{\{b_i = \max_{j \in N} b_j\}}}{|\{i \in N : b_i = \max_{j \in N} b_j\}|} \right)_{i \in N}.$$

And the expected transfer of any player is $\mu_i^I(b) = \pi_i b_i$ for FPA and $\mu_i^{II}(b) = \pi_i b^{(2)}$ for SPA, where $b^{(2)}$ stands for the 2nd-highest one in $b_1, \cdots, b_n$.

Thus, we may provide the setting of mechanism design based on auctions discussed above. Let $N = \{1, \cdots, n\}$ be the set of participants. We assign each participant a type $\theta_i \in \Theta_i$ and denote $\Theta = \prod_{i \in N} \Theta_i$.

Notice that mechanism is a generalized auction and the properties of auction may be a good starting point for understanding mechanisms. There are several important questions:

- Recall SPA, participants truthfully report their value. We may ask whether we can find mechanisms that nudge people to do truth-telling and why they would like to reveal their true type.
- Notice that people have incentive to join a FPA because their ex ante expected payoff is strictly higher than 0 according to (7). So why would participants join an arbitrary mechanism? We would like to know the criteria for voluntary participation.
- We've shown that mechanisms satisfying conditions in **theorem 7.3** provides the same revenue for the auctioneer. We would like to ask, whether there exists general criteria that can help distinguish mechanisms that yields the same revenue.
- Since FPA and SPA yields the same expected revenue and paymant, then the mechanism designer and participants should be indifferent between them. But what's the difference between them? That is to say, how can we compare between some mechanisms?

7.2.2 Direct Mechanism and Incentive Compatilibity

We first consider mechanisms where participants have incentive to reveal the true state. Suppose participents reveals the true state, then the mechanism designer may allocate the object according to their true type, i.e., commit to a allocation rule over the type space. Such mechanism is called a direct mechanism.

Definition 7.2 (Direct mechanism)**.** A direct mechanism is a 3-tuple $\langle q, t \rangle$, where $q : \Theta \to \Delta$ is

the probability of allocating the object to a certain participant and $t : \Theta \to \mathbb{R}^N$ is the expected transfer participants have to pay.

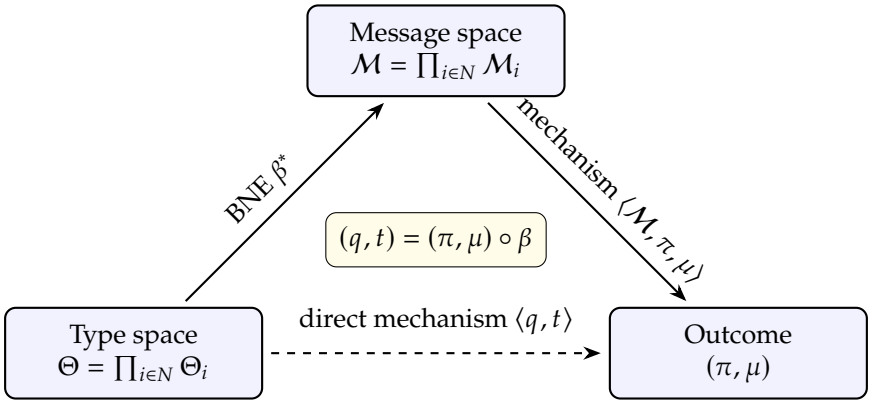


Figure 2: Mechanism, Direct Mechanism and Revealation Principle

The following result shows that we can always focus on a direct mechanism, i.e., every mechanism has an equivalent direct mechanism in the sense that they yield the same outcome.

Theorem 7.4 (Revealation principle)**.** *For an arbitrary mechanism $\langle \mathcal{M}, \pi, \mu \rangle$, there exists a direct mechanism $\langle q, t \rangle$ that yields the same outcome.*

Proof. See **figure 2**. □

Even though there always exists direct mechanisms, it does not necessarily mean that participants will truthfully report their type. To incentivize participants to do truth-telling, designer should design a mechanism that makes truth-telling a weakly dominant strategy for every participant in the interim stage. We define Q_i and T_i as the interim expected probability of winning the object and the interim expected transfer, i.e.,

$$Q_i(\theta_i) = \int_{\Theta_{-i}} q(\theta_i, \theta_{-i}) f_{-i}(\theta_{-i}) d\theta_{-i};$$
$$T_i(\theta_i) = \int_{\Theta_{-i}} t_i(\theta_i, \theta_{-i}) f_{-i}(\theta_{-i}) d\theta_{-i}.$$

Then the interim expected payoff U_i can be written as $\theta_i Q_i(\theta_i) - T_i(\theta_i)$.

To make $\beta_i(\theta_i)$ a weakly dominant strategy, we require

$$\forall \theta' \neq \theta_i \in \Theta_i : U_i(\theta_i) \geq \theta_i Q_i(\theta', \theta_{-i}) - T_i(\theta').$$

We call this condition incentive compatilibity.

Definition 7.3 (Incentive compatilibity)**.** A direct mechanism is incentive compatible(IC hereafter) if and only if for every player i, we have

$$\forall \theta' \neq \theta_i \in \Theta_i : U_i(\theta_i) \geq \theta_i Q_i(\theta', \theta_{-i}) - T_i(\theta').$$

We provide some properties of IC direct mechanisms.

Proposition 7.5. *Suppose $\langle q, t \rangle$ is a IC direct mechanism, then we have the followings:*

(1) Q_i and U_i are increasing;
(2) U_i is convex and not differentiable at at most countable points, hence $U_i' = Q_i$ a.w.;
(3) Payoff equivalence: $U_i(\theta_i) = U_i(\underline{\theta}) + \int_{\underline{\theta}}^{\theta_i} Q_i(x)\, dx$;
(4) Revenue equivalence: $T_i(\theta_i) = T_i(\underline{\theta}) + \int_{\underline{\theta}}^{\theta_i} x\, dQ_i(x)$

Proof. The proof is trivial. (2) is a standard result in real anaysis; for details, see Ok (2007, results of....). Then (3) is done and (4) can be obtained by substituting $U_i(\theta_i) = \theta_i Q_i(\theta_i) - T_i(\theta_i)$. □

We may provide a complete characterization of IC.

Theorem 7.6. *A direct mechanism $\langle q, t \rangle$ is IC if and only if the followings hold:*

(1) Q_i is increasing;
(2) $\forall \theta_i \in [\underline{\theta}, \bar{\theta}]$:

$$T_i(\theta_i) = T_i(\underline{\theta}) + \int_{\underline{\theta}}^{\theta_i} x\, dQ_i(x).$$

7.2.3 Individual Rationality

7.2.4 Optimal Mechanism

7.2.5 Efficient Mechanism

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